Test 7 – Grade 8

Name: _____
Grade / Class: 8

Section A: English Language & Literature

1. Reading Comprehension
Read the passage and answer the questions:

"During the school science exhibition, Leila displayed a model of the solar system. She explained how planets orbit the sun and the differences in size and composition. Visitors asked her questions about space exploration and astronomy careers. By the end of the day, Leila was thrilled to inspire her classmates with scientific knowledge."

a) What did Leila display at the exhibition?
b) Name two points Leila explained to visitors.
c) How did Leila feel after the event?
d) Summarize the passage in four sentences using your own words.

2. Grammar & Vocabulary
a) Combine these sentences using a suitable conjunction:
"The train was delayed. The passengers remained calm."

b) Identify all verbs in the sentence:
"The dedicated volunteers cleaned the park and planted saplings for the community."

c) Correct the error:
"Neither the teacher nor the students was ready for the sudden drill."

d) Change into reported speech:
"He said, 'I have completed my assignment ahead of the deadline.'"

3. Writing
Write a short essay (120–150 words) on the topic: "Advantages and Disadvantages of Online Learning". Include challenges, benefits, and your opinion.

Section B: Mathematics

1. Algebra
a) Solve for x:
$6x + 14 = 4x + 30$

b) Factorize:
$x^2 – 13x + 36$

c) Simplify:
$8a^2b – 5ab^2 + 3a^2b$

2. Geometry & Mensuration
a) The sides of a rectangle are 18 cm and 12 cm. Find its area and perimeter.
b) A cone has radius 7 cm and height 24 cm. Find its volume.

3. Ratio, Proportion & Percentage
a) A jacket originally costs $120. It is sold at a 20% discount. Find the selling price.
b) If 12 notebooks cost $48, find the cost of 25 notebooks.

4. Data Handling
The scores of 12 students in a geography test: 14, 16, 18, 17, 19, 16, 15, 17, 18, 16, 15, 19.
a) Calculate the mean, median, and mode.
b) Represent the data in a frequency table.

Section C: Science

1. Biology
a) What are the main functions of the human heart?
b) Explain the process of respiration in plants.
c) State two differences between amphibians and fish.

2. Physics
a) A cyclist accelerates from 5 m/s to 15 m/s in 10 seconds. Calculate acceleration.
b) Define work and give an example.
c) Explain the difference between reflection and refraction of light.

3. Chemistry
a) Classify the following as element, compound, or mixture:

- **Hydrogen**
- **Carbon dioxide**
- **Saltwater**

b) Write a balanced chemical equation for the reaction between magnesium and hydrochloric acid.

4. Environmental Science
a) List two soil pollutants and their impact.
b) Suggest three practical methods to reduce air pollution in urban areas.

Section D: Social Studies

1. History
a) Name two major revolutions of the 19th century and their effects.
b) Explain the importance of the Renaissance in Europe.
c) Identify the members of the Axis Powers in World War II.

2. Geography
a) Define equator and explain its effect on climate.
b) Name the largest desert in Africa and the continent it is located in.
c) Discuss two consequences of urbanization on the environment.

3. Civics / Political Science
a) Differentiate between democratic and autocratic forms of government.
b) State two rights of citizens in a democracy.

4. Economics
a) Explain demand and supply using a real-life example.
b) Differentiate between short-term and long-term business costs.

Section E: ICT / Computer Studies

1. Computer Basics
a) What is the internet? Give one common use.
b) Name one input device and one output device.

2. Applications
a) Which spreadsheet function calculates the smallest value in a dataset?
b) Name two uses of word processing software in schools.

3. Programming Basics
a) Define a loop in programming and give an example.
b) Write a Python snippet to print all odd numbers from 1 to 25.

4. Cyber Awareness
a) What is ransomware and how can it be prevented?
b) List two ways to protect personal information on social media.

Section F: Islamiyat

1. Quran & Hadith
a) Mention two moral lessons taught in the Quran.
b) Narrate a Hadith that emphasizes kindness to neighbors.

2. Aqidah / Belief
a) List the five pillars of Islam.
b) Explain the importance of belief in the oneness of Allah.

3. Fiqh / Worship
a) Why is fasting during Ramadan obligatory?
b) Describe the spiritual significance of giving Zakat.

4. Islamic Ethics & Morality
a) How does Islam encourage fairness in trade?
b) Explain the importance of environmental stewardship in Islam.

Section G: Arts & Music

1. Visual Arts
a) Explain how symmetry is used in art.
b) Name two secondary colors.

2. Music
a) Define pitch and volume in music.
b) Give an example of a wind instrument.

Section H: Health & Physical Education

1. Physical Fitness
a) Name two exercises for improving cardiovascular health.
b) List one recreational activity and one competitive sport.

2. Health & Hygiene
a) Why is maintaining hydration important?
b) Name two ways to prevent food-borne illnesses.

3. Safety & Environment
a) State one safety rule for using electrical appliances.